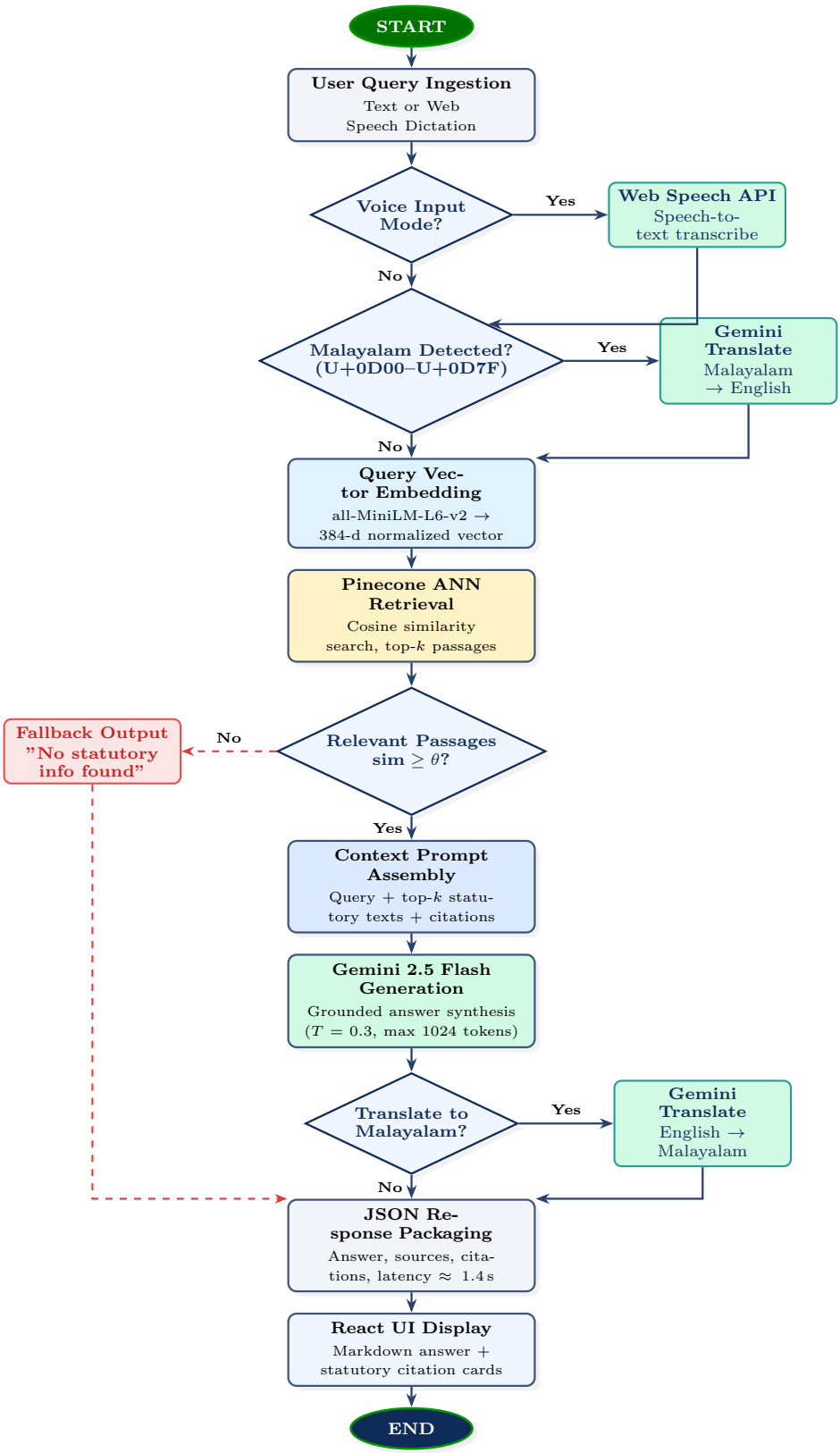

START
User Query Ingestion
Text or Web Speech Dictation
Voice Input Mode?
Yes
Web Speech API
Speech-to-text transcribe
No
Malayalam Detected? (U+0D00–U+0D7F)
Yes
Gemini Translate
Malayalam → English
No
Query Vector Embedding
all-MiniLM-L6-v2 → 384-d normalized vector
Pinecone ANN Retrieval
Cosine similarity search, top-k passages
Relevant Passages sim ≥ θ?
No
Fallback Output "No statutory info found"
Yes
Context Prompt Assembly
Query + top-k statutory texts + citations
Gemini 2.5 Flash Generation
Grounded answer synthesis (T = 0.3, max 1024 tokens)
Translate to Malayalam?
Yes
Gemini Translate
English → Malayalam
No
JSON Response Packaging
Answer, sources, citations, latency ≈ 1.4 s
React UI Display
Markdown answer + statutory citation cards
END